EC23232 – DIGITAL LOGIC AND MICROPROCESSOR

UNIT I - MINIMIZATION TECHNIQUES AND LOGIC GATES

PART–A

1. Write De-Morgan’s Theorem.
2. Write duality Principle in Boolean Algebra.
3. Derive the simplified form of AB + A’C + BC using Boolean Laws.
4. Convert the decimal number 348.35 into hexadecimal.
5. List the non weighted binary codes.
6. Convert the binary number 1100 into gray code.
7. Which gates are called as Universal gates? Why are they called so?
8. Implement AND gate using NAND gate.
9. Write the expression and truth table of EX-OR gate.
10. Perform 2’s complement arithmetic of 9-12.
11. Show that A + A’B + AB’ = A + B using Boolean simplifications.
12. State and prove the Involution Law in Boolean Algebra.
13. Find the complement of the function F= (AB)’ + (C + D)’
14. Convert the POS into canonical form F=(A+B)(B+C)
15. How many NOR gates are required to implement AND gate and OR gate?

PART–B

1. Reduce using K map F(A, B, C, D) = Σm (0, 1, 2, 5, 7, 8, 9, 10, 13, 15) and implement using universal gate.
2. Reduce the given function F(A,B,C,D) = Σ (1,2,3,5,8,10,11,12) using K-map and implement using universal logic.
3. Simplify the SOP function F(w,x,y,z) = Σ (4,5,6,7,12,13,14,15) using K-map and implement using NAND gate.
4. Reduce F(A,B,C,D) = ∏(0,1,2,4,5,7,10,15) using K-map and implement using NOR gate.
5. Simplify the following in (i) SOP (ii) POS and implement in basic gates F(A, B, C, D) = Σm (0,1,2,5,8,9,10)
6. Minimize the following 4-variable Boolean expression in SOP form using K-map (A,B,C,D)=∑m(0,1,4,5,6,10,13)+d(2,3)
7. Minimize the following 4-variable Boolean expression in POS form using K map. F(A,B,C,D)=∏M(1,5,6,12,13,14)+d(2,4)
8. Simplify using tabulation method : F(A,B,C,D) =∑ m(0,1,2,4,6,8,9,11,13,15)
9. Simplify the following expression to SOP using tabulation method : F(A,B,C,D) =∑ m(0,1,2,3,4,6,7,,11,12,15)
10. Simplify the expression using tabulation method : F(A,B,C,D) =∑ m(0,4,8,10,12,13,15)+ d(1,2)

UNIT II -COMBINATIONAL AND SEQUENTIAL CIRCUITS

PART–A

1.What is a combinational logic circuit?
2. What is priority Encoder?
3. Give the applications of De-multiplexer.
4. Write an Expression for borrow and difference in a half Subtractor circuit.
5. Draw the logic diagram of full adder.
6. What is data distributor and why it is called so?
7. Is that possible to use Demultiplexer as decoder? Justify your answer.
8. What is the drawback of parallel adder?
9. Define Flip flop.
10. Define sequential circuit.
11. Compare combinational and sequential circuits.
12. How many flip-flops are required to count 120 clock pulses?
13. What is the drawback of SR flip-flop and how it can be overcome?
14. Distinguish between synchronous and asynchronous counter.
15. What are the different types of shift registers?

PART–B

1. Design half adder and full adder circuit.
1. Implement the following function using 8 to 1 Multiplexer.

$$F(A,B,C,D) = \Sigma\ (0,1,3,4,12,14,15)\ )$$

2. Design a 2 bit magnitude comparator and explain its operation in detail.
3. Realize a Binary to gray code conversion circuit.
4. Design a priority encoder with suitable logic gates.
5. Design and explain the operation of 4 bit parallel adder/ subtractor.
6. Explain the operation of clocked JK flip-flops with suitable diagrams.
7. Design a synchronous counter with states 0,1, 2,3,0,1 …………. using JK flip-flop.
8. Design Mod-6 counter using T flip-flop.
9. Explain the different shift register with neat diagrams.
10. Design 3 bit UP/DOWN counter using JK flip-flops.

UNIT III - THE 8085 MICROPROCESSOR

PART A

1. Difference between microprocessor and microcontroller.
2. How many **address lines and data lines are there in** 8085 microprocessor.
3. Why status signals are provided in microprocessor?
4. How is the **program counter (PC) used in memory addressing?**
5. List the Software and Hardware interrupts of 8085.
6. What is vectored and Non- Vectored interrupt?
7. What is masking and why it is required?
8. List the flags of 8085.
9. What is ALE?
10. Define machine cycle.
11. Write the instructions to add two numbers $F2_H$ and $F8_H$ and store the result in 4300_H
12. Compare JMP and CALL instruction of 8085.
13. List the addressing modes of 8085 microprocessor.
14. What is the function performed by EI and Dl instruction?
15. What is the role of the Accumulator register in the 8085?

PART B

1. Explain the architecture of 8051 microcontroller.
2. Draw the pin diagram of 8051 microcontroller and explain the functionality of each pin in detail.
3. Explain the different addressing modes of 8051 with examples.
4. Explain the classification of instructions sets of 8051microcontroller with examples.
5. Develop assembly language programming to add and subtract two 8 bit numbers.
6. Develop assembly language programming to implement 8 bit multiplication and 8 bit division.
7. Describe the modes of operation of timers in the 8051.
8. Describe the Serial Control (SCON) register used in UART communication.
9. Explain 8051 serial port programming with examples.
10. Explain the interrupt structure of 8051 microcontroller Explain how interrupts are prioritized.

UNIT IV - THE 8051 MICROCONTROLLER

PART A

16. Distinguish between microprocessor and Microcontroller.
17. List the features of 8051 microcontroller.
18. What are special functions registers?
19. Distinguish between Harvard and Von Neumann architecture.
20. List out the addressing modes of 8051.
21. Write the PSW format of 8051 microcontroller.
22. Compare unconditional jump instructions.
23. Name the five interrupt sources of 8051.
24. Mention the register banks and their addresses in 8051.
25. Explain the two power saving modes of operation of 8051.
26. Specify the function of ALE in 8085 Microprocessor.
27. What are the functions of PUSH and POP instruction in a 8051 Microcontroller
28. Define interrupt.
29. Which register is used for serial programming in 8051 microcontroller?
30. Specify the number of bits of count that can be used in different modes of timer/counter of 8051.

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UNIT V – INTERFACING AND APPLICATIONS

PART A

1. How is handshaking done in Mode 1 and Mode 2 of the 8255?
2. What is the function of the control register in the 8255?
3. List the different operating modes of the 8255?
4. Compare 8253 and 8254 timers.
5. What is the function of the Mode Control Register in the 8253?
6. Write the different operating modes of the 8253 timer.
7. How does the 8253 handle frequency division and pulse generation?
8. Why are ADCs and DACs required in microprocessor-based systems?
9. What is the function of the START and EOC (End of Conversion) signals in ADC?
10. Specify the function of the Sampling and Hold Circuit in ADC interfacing.
11. Write the function of a Latch in DAC interfacing.
12. Compare Full Step, Half Step, and Micro stepping modes.
13. How number of steps per revolution is calculated in a stepper motor?
14. List the applications of stepper motor.
15. Write the role of an encoder in stepper motor systems.

PART B

1. Explain the architecture of the 8255 Programmable Peripheral Interface with a neat block diagram.
2. Describe the control word format and modes of operation of 8255 Programmable Peripheral Interface.
3. Explain the architecture of the Programmable Interval Timer 8253 timer with a neat block diagram.
4. Describe the different operating modes of the 8253 timer with the control word register.
5. Explain the interfacing diagram of ADC 0804 with 8051 microcontroller.
6. Brief the interfacing of DAC 1408 with 8051 microcontroller with a neat sketch.
7. Design a microcontroller system for traffic light control.
8. Describe how an 8051 microcontroller can control a stepper motor.
9. Describe how the direction and speed of a stepper motor is controlled with microcontroller.
10. Explain different phases and cycles in a traffic light control system.